Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

Learning goal: I can apply the laws of exponents with whole-number exponents. (circle your level after scoring)

4	I can combine several exponent rules in one expression and explain each step.
3	I can use the product, quotient, and power rules to write an equivalent expression.
2	I can evaluate a power with a whole-number exponent.
1	I need help to start this benchmark.

1. Evaluate the expression. Write your answer as a fraction in simplest form.

$$\left(-\frac{2}{3}\right)^3 \times \left(-\frac{2}{3}\right)^3$$

2. Which expression is equivalent to $4^5 \times 4^2$?

A. 4^7

B. 16^7

C. 4^3

D. 4^{10}

3. Evaluate the expression.

$$(-10)^6 \div (-10)^4$$

Turn over →

4. The equation shown is true.

$$(6^4)^n = 6^{16}$$

What is the value of n?

5. A student wrote $\left(\left(\frac{1}{2}\right)^5\right)^3 \div \left(\frac{1}{2}\right)^4$ as a single power.

Step 1: $\left(\frac{1}{2}\right)^8 \div \left(\frac{1}{2}\right)^4$

Step 2: $\left(\frac{1}{2}\right)^4$

In which step did the student make the first error? Write the expression correctly as a single power.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-NSO.1.1 · MA.7.NSO.1.1 · Laws of exponents (whole-number exponents)

Unit 8: Evaluating Numeric Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $\frac{64}{729}$

Multiplying powers with the same base adds the exponents: $3 + 3 = 6$, so the expression equals $\left(-\frac{2}{3}\right)^6$.

$\left(-\frac{2}{3}\right)^6 = \frac{64}{729}$. A negative base to an even power is positive.

2. Regular **A.** 4^7

When multiplying powers with the same base, add the exponents.

$4^5 \times 4^2 = 4^7$.

3. Regular **100**

The bases are the same, so subtract the exponents: $6 - 4 = 2$.

$(-10)^2 = 100$.

A negative base raised to an even power is positive. Only the exponents change — the base stays −10.

4. Regular **4**

A power raised to a power multiplies the exponents, so $4 \times n = 16$.

$n = 16 \div 4 = 4$.

5. Rigor **Step 1; $\left(\frac{1}{2}\right)^{11}$. Score: 1 point for the step and the correct single power; the step alone earns 0.**

Step 1 adds 5 and 3; a power raised to a power multiplies them: $5 \times 3 = 15$.

Correct work: $\left(\left(\frac{1}{2}\right)^5\right)^3 = \left(\frac{1}{2}\right)^{15}$, and $\left(\frac{1}{2}\right)^{15} \div \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^{11}$.

Re-check strip (3 items, C-NSO.1.1-R)

1. Regular **243**

Multiplying powers with the same base adds the exponents: $3 + 2 = 5$, so the expression equals 3^5.

$3^5 = 243$.

2. Regular 9^8

A power raised to a power multiplies the exponents: $4 \times 2 = 8$.

So $(9^4)^2 = 9^8$.

Adding the exponents here would give 9^6; exponents are added only when powers are multiplied.

3. Regular **1000**

Dividing powers with the same base subtracts the exponents: $7 - 4 = 3$.

$10^3 = 1000$.

Only the exponents change — the base stays 10, it is not divided.

Comparison strip (2 items, C-NSO.1.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **5**

Write 8 as a power of 2: $8 = 2^3$.

Dividing powers with the same base subtracts the exponents, so $n - 2 = 3$.

$n = 3 + 2 = 5$.

2. Regular **625**

Dividing powers with the same base subtracts the exponents: $7 - 3 = 4$.

$5^4 = 625$.

Only the exponents change — the base stays 5, it is not divided.

Reteach plan — C-NSO.1.1

MA.7.NSO.1.1 · Laws of exponents (whole-number exponents)

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-NSO.1.1-R

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. Evaluate the expression.

$3^3 \times 3^2$

2. Write the expression as a single power.

$(9^4)^2$

3. Evaluate the expression.

$10^7 \div 10^4$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-NSO.1.1-R

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. Evaluate the expression.

$3^3 \times 3^2$

2. Write the expression as a single power.

$(9^4)^2$

3. Evaluate the expression.

$10^7 \div 10^4$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2) — C-NSO.1.1-X

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. The equation shown is true.

$2^n \div 2^2 = 8$

What is the value of n?

2. Evaluate the expression.

$5^7 \div 5^3$

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2) — C-NSO.1.1-X

Benchmark: **MA.7.NSO.1.1** · Laws of exponents (whole-number exponents) Student code: _____ Calculator: no

1. The equation shown is true.

$2^n \div 2^2 = 8$

What is the value of n?

2. Evaluate the expression.

$5^7 \div 5^3$

Score: ____ / 2 → **copy to your progress strip**